III SEM. B.TECH.MID-TERM EXAMINATION (CSE/CCE/IT)

ENGINEERING MATHEMATICS III (MAT 2126) Time: 20 mins

Q1. If *G* is a graph on 7 vertices such that all its vertices have degree 4, then the number of edges of the complement of *G* is ... (0.5)

1. 7
2. 6
3. 9
4. 21

Q2. Let G be a simple connected graph with 5 vertices. The degree of 4 vertices are 4,2,2,1. Then the fifth vertex has degree equal to … (0.5)

1. 0
2. 2
3. 3
4. 5

Q3. The maximum number of edges in a bipartite graph on 14 vertices is...(0.5)

1. 56
2. 14
3. 49
4. 87

Q4. The number of vertices required to construct a simple connected graph with 6 edges in which each vertex is of degree 2 is… (0.5)

1. 6
2. 12
3. 2
4. 4

Q5. The total number of nonzero entries in the adjacency matrix of any tree on n vertices is… (0.5)

1. 2(n-1)
2. n-1
3. $\frac{n-1}{2}$
4. n

Q6. Let G be a complete graph on n vertices, then…(0.5)

1. G is Eulerian for every natural number
2. G is Eulerian only when n is even
3. G is Eulerian only when $\boldsymbol{n}$ is odd
4. G is never Eulerian

Q7. Define $*$ on Q^+(Set of positive rational numbers without zero) as $a * b = \frac{ab}{2}$, for all $a, b \in Q^+$. Then $(Q^+, *)$ is a group. The inverse of $2 \in Q^+$ is… (0.5)

1. $\frac{1}{4}$
2. $\frac{1}{2}$
3. 2
4. 1

Q8. Which of the following cannot be the order of subgroup of a group with 12 elements… (0.5)

1. 1
2. 2
3. 4
4. 8

Q9. Let G be a group of order 15 and let H be a subgroup of G of order 3. Then the number of distinct left cosets of H in G is … (0.5)

1. 2
2. 3
3. 4
4. 5

Q10. Which of the following is necessarily true in a group G? (0.5)

1. $(ab)^{-1} = a^{-1}b^{-1}$ for all $a, b \in G$
2. $(ab)^{-1} = b^{-1}a^{-1}$ for all $a, b \in G$
3. $(ab)^2 = a^2b^2$ for all $a, b \in G$
4. $(ab)^2 = b^2a^2$ for all $a, b \in G$

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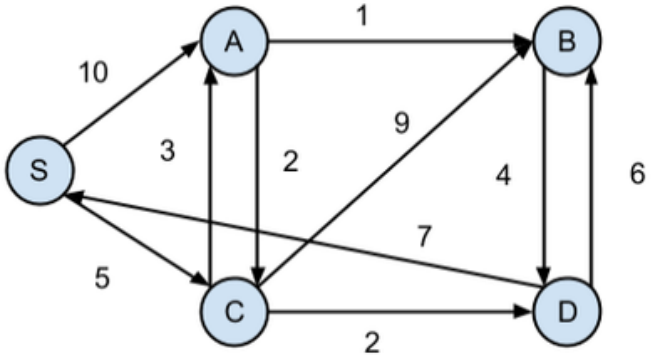
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Answer all questions.

Q11. Show that a connected graph is bipartite if and only if it has no odd cycles. Give an example of a bipartite graph. (4 Marks)

Q12. Prove that a nonempty subset H of a group $(G,*)$ is a subgroup of G if and only if $a * b^{-1} \epsilon H$ for all $a, b \epsilon H$. Find all subgroups of $\mathbb{Z}_4$ under addition modulo 4. (4 Marks)

Q13. Using Dijkstra's algorithm, find the shortest paths from S to all the other vertices.



(3 Marks)

Q14. If G is a graph, then show that either G or $\bar{G}$ is connected. Draw a self complementary graph and find its diameter. (3 Marks)

Q15. A tree has 2n vertices of degree 1, 3n vertices of degree 2 and n vertices of degree 3. Determine the number of vertices and the number of edges in the tree.

(3 Marks)

Q16. Let H be a subgroup of G. Show that any two distinct left cosets of H in G are disjoint. (3 Marks)

Q17. Prove that $M = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} | a, b, c, d \text{ are real numbers} \right\}$ forms a group with respect to matrix addition. Is this group is Abelian? (3 Marks)

Q18. Let a and b be elements in a group G. Using Mathematical induction, prove that $ab^n a^{-1} = (aba^{-1})^n$ for any natural number n. (2 Marks)